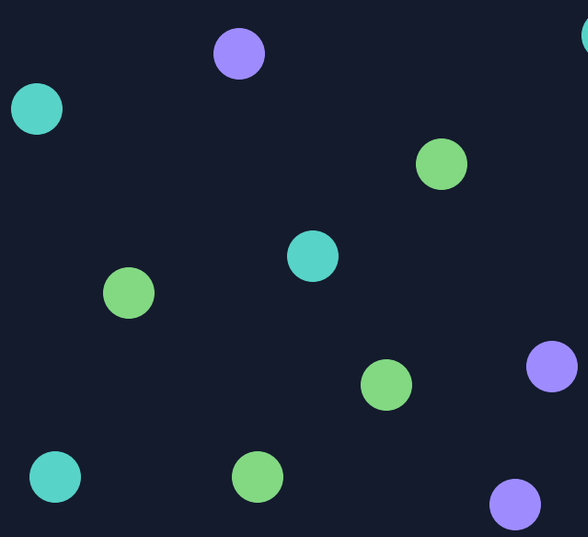
VEC FAILURE MODE GALLERY

# Population collapse

Many generated cells become nearly identical.

HEALTHY



FAILED



**LIKELY SYMPTOM**

MMD usually worsens; co-variation may also degrade.

**DEBUG NEXT**

Check duplicate/near-duplicate cells, per-gene variance, effective rank.



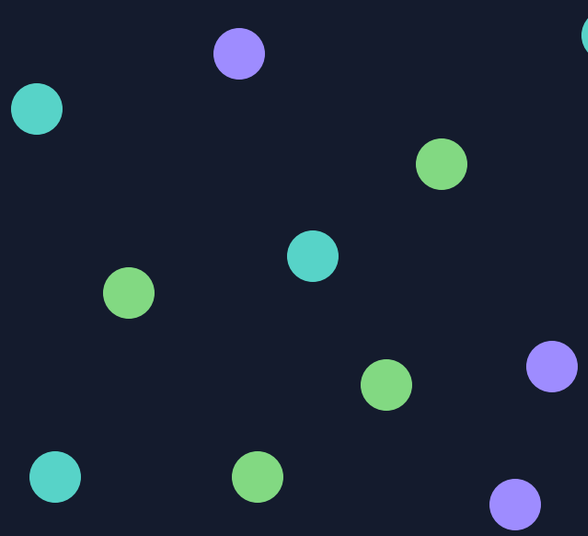
Community visual resource - MIT

VEC FAILURE MODE GALLERY

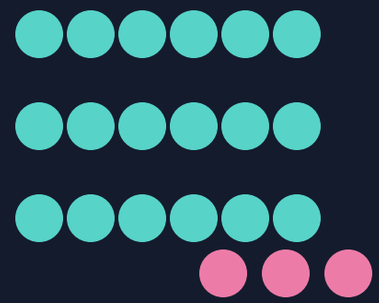
# Wrong cell-state mixture

States exist, but their proportions are wrong.

HEALTHY



FAILED



**LIKELY SYMPTOM**

MMD is the clearest population-level warning.

**DEBUG NEXT**

Compare embeddings/composition; avoid relying on pseudobulk alone.



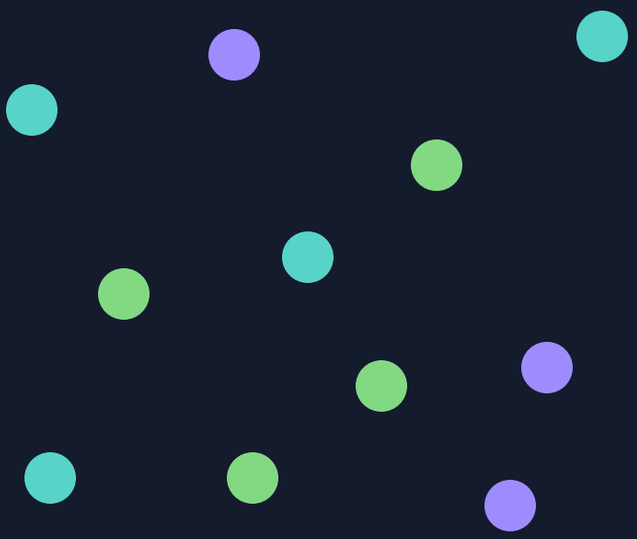
Community visual resource - MIT

VEC FAILURE MODE GALLERY

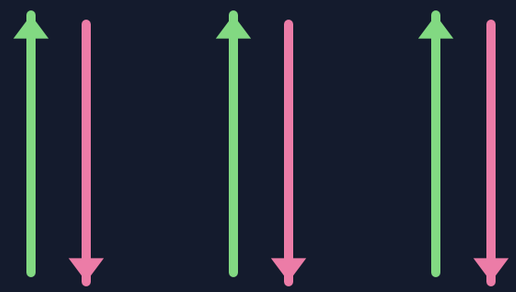
# Reversed gene response

The model moves responsive genes in the wrong direction.

HEALTHY



FAILED



**up <-> down**

**LIKELY SYMPTOM**

DCS / de_direction falls; DES can still look partly reasonable.

**DEBUG NEXT**

Plot predicted vs observed/reference change; inspect sign agreement.



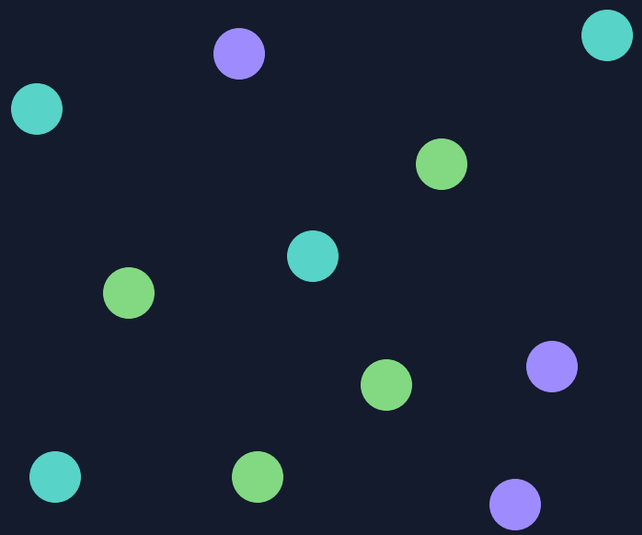
Community visual resource - MIT

VEC FAILURE MODE GALLERY

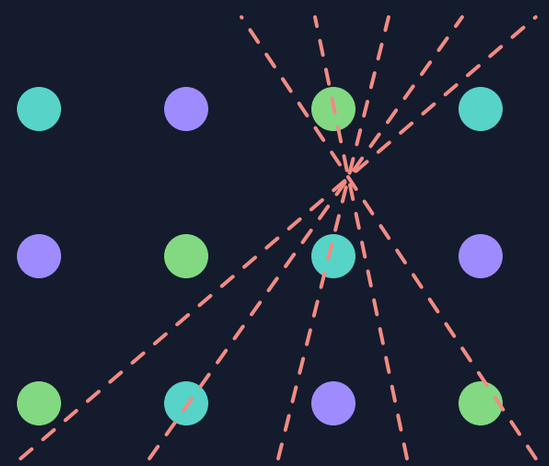
# Scrambled co-expression

Per-gene distributions look right, but genes no longer vary together.

HEALTHY



FAILED



**LIKELY SYMPTOM**

CSS / variogram worsens while marginal metrics may remain decent.

**DEBUG NEXT**

Inspect covariance or co-expression structure across cells.



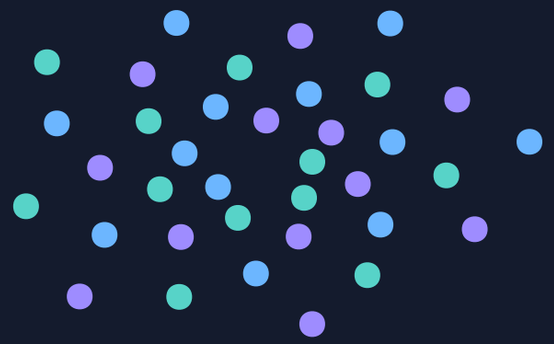
Community visual resource - MIT

VEC FAILURE MODE GALLERY

# Spatial line collapse

A 3D tissue collapses toward one dimension.

HEALTHY



FAILED



**LIKELY SYMPTOM**

T2 shape terms degrade strongly; local organization may also suffer.

**DEBUG NEXT**

Inspect geometric rank, PCA axes, duplicate coordinates.



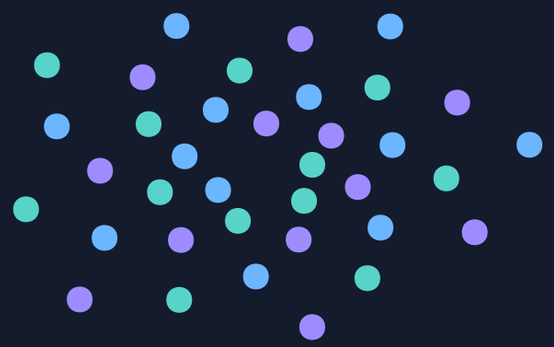
Community visual resource - MIT

VEC FAILURE MODE GALLERY

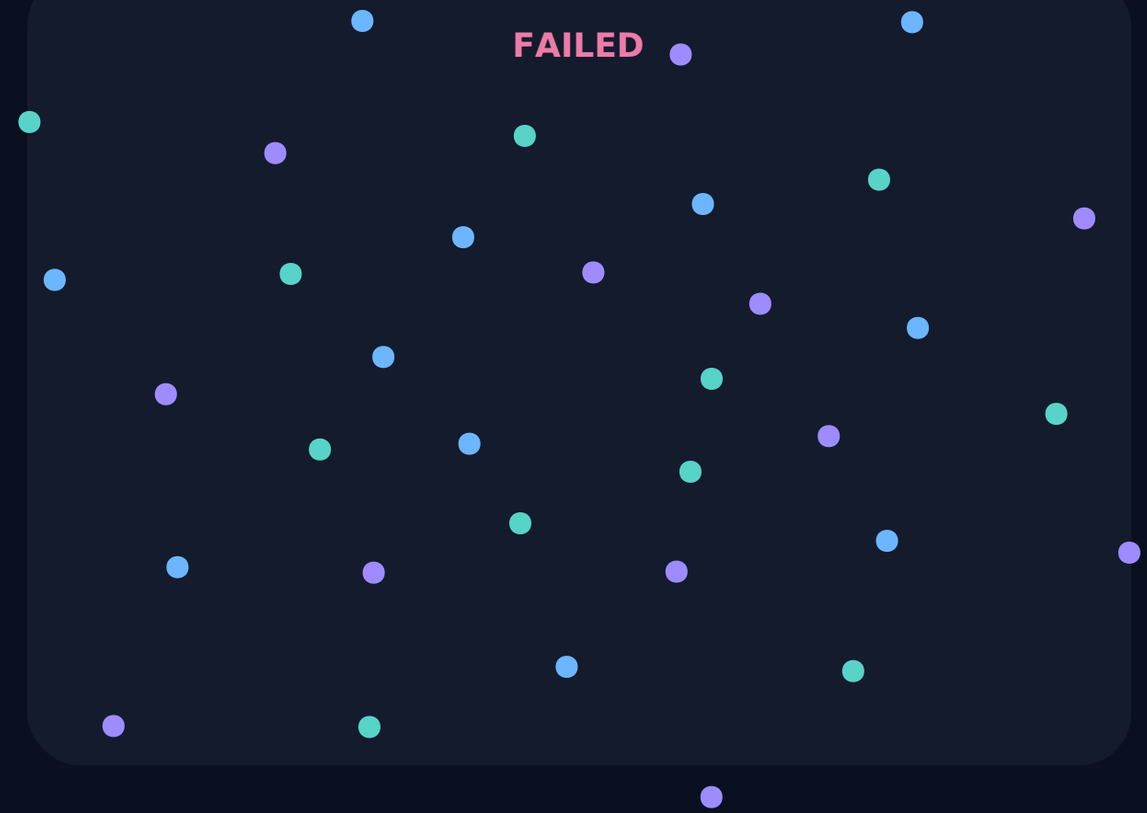
# Uniform scale error

The predicted tissue shape is plausible but globally too large/small.

HEALTHY



FAILED



**LIKELY SYMPTOM**

TSR / scale_log_ratio moves away from zero; shape may otherwise look good.

**DEBUG NEXT**

Compare RMS radius and pairwise-distance scale before altering topology.



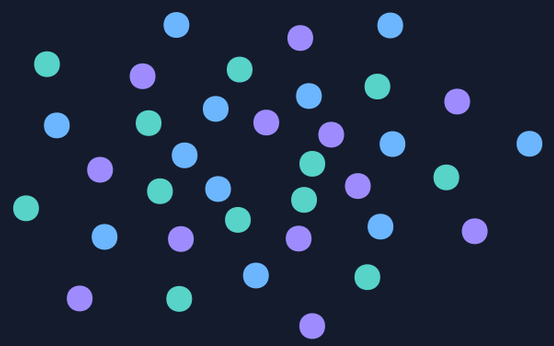
Community visual resource - MIT

VEC FAILURE MODE GALLERY

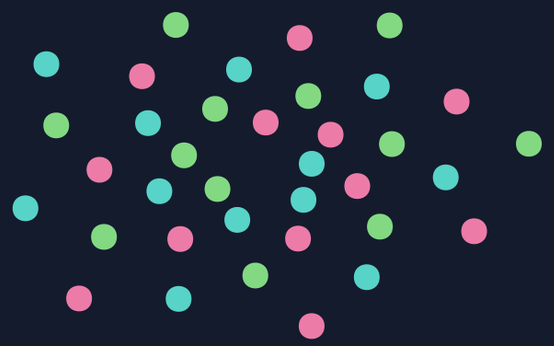
# Neighborhood scramble

Global geometry and cell states look right, but states are assigned to wrong locations.

HEALTHY



FAILED



**LIKELY SYMPTOM**

NFS / neighborhood_mmd worsens; global shape can stay good

**DEBUG NEXT**

nspect local neighbors and expression-position coupling.



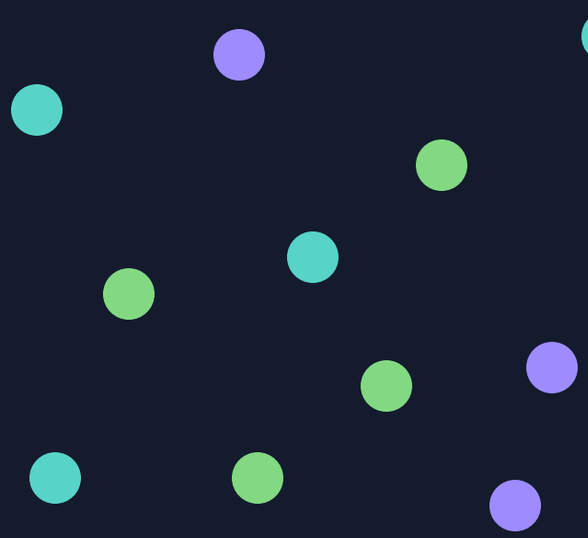
Community visual resource - MIT

VEC FAILURE MODE GALLERY

# Raw-count expression

A structurally valid file is on the wrong expression scale.

HEALTHY



FAILED

0 1 0 7 3

12 0 0 4 28

**integer-like counts**

instead of log scale

**LIKELY SYMPTOM**

Many metrics can fail unpredictably; format validation may still pass.

**DEBUG NEXT**

Check integer fraction, value range, per-cell sums; submit log-normalized values.



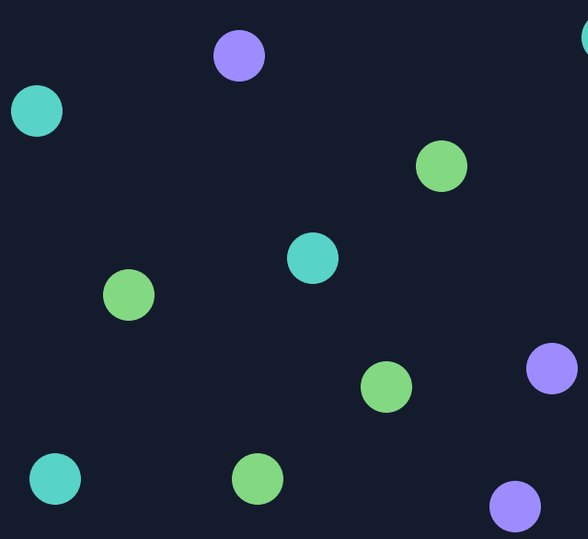
Community visual resource - MIT

VEC FAILURE MODE GALLERY

# Gene-order mismatch

The right genes are stored in the wrong column order.

HEALTHY



FAILED

Nkx2-5

Actb

Gata4

Tbx5

**same set - wrong order**

**LIKELY SYMPTOM**

Usually a validation error, not a low biological score.

**DEBUG NEXT**

Reindex columns to the released ordered panel / transcriptome.



Community visual resource - MIT